KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3382 | 3 Credits: 4 | Program Core | Source-grounded

Introduction to Artificial

Students will be able to summarize the fundamental concepts of artificial intelligence and

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3382 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3382.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Artificial Intelligence
Course code	3382
Course title	Introduction to Artificial
Semester	3 Credits: 4
Course category	Program Core
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

21 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Students will be able to summarize the fundamental concepts of artificial intelligence and
- solve various problems using AI techniques.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding the significance of Intelligent Agents in AI. Apply Mathematical logic for knowledge	See official syllabus
CO2	15 Applying representation of AI	See official syllabus
CO3	Practice various search techniques in AI 16 Applying	See official syllabus
CO4	Describe learning and expert systems. 14 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Intelligent Agents in AI.	7	As specified
Module 2	Apply Mathematical logic for knowledge representation of AI	7	As specified
Module 3	Practice various search techniques in AI	3	As specified
Module 4	Familiarize with constraint satisfaction problems in AI	4	As specified

MODULE 1

Intelligent Agents in AI.

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Intelligent Agents in AI.
- Official duration: As specified
- Official topic entries captured: 7

- The full source excerpt is preserved below for coverage checking.

Define AI, foundation and history of AI.

Define AI, foundation and history of AI. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Define AI, foundation and history of AI. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, define ai, foundation and history of ai. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Define AI, foundation and history of AI. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Define AI, foundation and history of AI. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Discuss applications of AI.

Discuss applications of AI. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discuss applications of AI. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discuss applications of ai. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discuss applications of AI. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Discuss applications of AI. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the Turing test.

Explain the Turing test. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the Turing test. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the turing test. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the Turing test. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the Turing test. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain AI problems

Explain AI problems is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain AI problems using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain ai problems should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain AI problems means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain AI problems [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain agents and environments.

Explain agents and environments. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain agents and environments. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain agents and environments. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain agents and environments. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain agents and environments. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Identify structure of agents

Identify structure of agents is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify structure of agents using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify structure of agents should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify structure of agents means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Identify structure of agents □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.

Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. is an official topic in Module 1 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, recognize types of agents. 2 understanding introduction to ai-ai concepts, foundations of ai, applications of ai -language models - information retrieval -

information extraction - natural language processing - machine translation - speech recognition, turing test. ai problems: water jug problem, tower of hanoi problem, four-queens problem, travelling salesman problem intelligent agents-agents and environments, structure of agent - types of agents. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Recognize types of agents. 2 Understanding Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language models - Information retrieval - Information extraction - Natural language processing - Machine translation - Speech recognition, Turing test. AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem, Travelling Salesman problem Intelligent Agents-Agents and Environments, Structure of Agent - types of agents. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

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Intelligent Agents in AI.

  M1.01      Define AI, foundation and history of AI.          2
Understanding
  M1.02     Discuss applications of AI.                        3
Understanding
  M1.03     Explain the Turing test.                           1
Understanding
  M1.04     Explain AI problems                                3
Understanding
  M1.05     Explain agents and environments.                   2
Understanding
  M1.06     Identify structure of agents                       2
Understanding
  M1.07     Recognize types of agents.                         2
Understanding

Contents:

Introduction to AI-AI concepts, Foundations of AI, Applications of AI -Language
models
- Information retrieval - Information extraction - Natural language processing -
Machine
translation - Speech recognition, Turing test.
AI Problems: Water Jug problem, Tower of Hanoi problem, Four-Queens problem,
Travelling Salesman problem
Intelligent Agents-Agents and Environments, Structure of Agent - types of agents.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Apply Mathematical logic for knowledge representation of AI

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Apply Mathematical logic for knowledge representation of AI
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

knowledge representation and reasoning Understanding Describe the representation and reasoning 2

knowledge representation and reasoning Understanding Describe the representation and reasoning 2 is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify knowledge representation and reasoning Understanding Describe the representation and reasoning 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, knowledge representation and reasoning understanding describe the representation and reasoning 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What knowledge representation and reasoning Understanding Describe the representation and reasoning 2 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] knowledge representation and reasoning Understanding Describe the representation and reasoning 2 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

about objects, relations, events, actions, time Understanding and space

about objects, relations, events, actions, time Understanding and space is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify about objects, relations, events, actions, time Understanding and space using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, about objects, relations, events, actions, time understanding and space should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What about objects, relations, events, actions, time Understanding and space means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ about objects, relations, events, actions, time Understanding and space □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Demonstrate Propositional Logic

Demonstrate Propositional Logic is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate Propositional Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate propositional logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate Propositional Logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Demonstrate Propositional Logic [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Demonstrate First Order Logic

Demonstrate First Order Logic is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Demonstrate First Order Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, demonstrate first order logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Demonstrate First Order Logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Demonstrate First Order Logic □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Explain Soundness and Completeness,

Explain Soundness and Completeness, is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Soundness and Completeness, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain soundness and completeness, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Soundness and Completeness, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain Soundness and Completeness, [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain Forward and Backward chaining

Explain Forward and Backward chaining is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Forward and Backward chaining using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain forward and backward chaining should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain Forward and Backward chaining means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain Forward and Backward chaining [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs.

Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. is an official topic in Module 2 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarize with classical planning 2 understanding knowledge representation and reasoning-ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. logic and inferences: propositional logic, first order logic, knowledge representation in first order logic, inference in first order logic, soundness and completeness, forward and backward chaining. classical planning - algorithms for planning state space search, planning graphs. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Familiarize with classical planning 2 Understanding Knowledge representation and reasoning-Ontologies, foundations of knowledge representation and reasoning, representing and reasoning about objects, relations, events, actions, time and space. Logic and Inferences: Propositional Logic, First Order Logic, Knowledge representation in First Order Logic, Inference in First Order Logic, Soundness and Completeness, Forward and Backward chaining. Classical Planning - Algorithms for planning state space search, Planning Graphs. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Apply Mathematical logic for knowledge representation of AI

            Summarize Ontologi es, foundations of              3
  M2.01     knowledge representation and reasoning
Understanding
            Describe the representation and reasoning            2
  M2.02     about objects, relations, events, actions, time
Understanding
            and space

  M2.03     Demonstrate Propositional Logic                      2
Applying

  M2.04     Demonstrate First Order Logic                        2
Applying

  M2.05     Explain Soundness and Completeness,                  2
Understanding
  M2.06     Explain Forward and Backward chaining                2
Understanding
  M2.07     Familiarize with classical planning                  2
Understanding
            Series Test – I                                      1
Contents:
Knowledge representation and reasoning-Ontologies, foundations of knowledge
representation and reasoning, representing and reasoning about objects,
relations, events,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Practice various search techniques in AI

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Practice various search techniques in AI
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth

Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth is an official topic in Module 3 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarise types of search algorithms 2 understanding implement breadth first search and depth should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

6 Applying First Search Implement Best first search, Hill Climbing, A*

6 Applying First Search Implement Best first search, Hill Climbing, A* is an official topic in Module 3 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying First Search Implement Best first search, Hill Climbing, A* using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying first search implement best first search, hill climbing, a* should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying First Search Implement Best first search, Hill Climbing, A* means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 6 Applying First Search Implement Best first search, Hill Climbing, A* □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search

6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search is an official topic in Module 3 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying & beam search search algorithm terminologies, types of search algorithms uninformed search: breadth first search, depth first search informed search: heuristic search - best first search, hill climbing, a* algorithm beam search should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Practice various search techniques in AI

  M3.01     Summarise types of search algorithms                 2
Understanding

            Implement Breadth First Search and Depth
  M3.02                                                          6       Applying
            First Search
            Implement Best first search, Hill Climbing, A*
  M3.03                                                          6       Applying
            & Beam Search
Contents:
Search Algorithm Terminologies, Types of search algorithms
Uninformed Search: Breadth First Search, Depth First Search
Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm
Beam Search
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Familiarize with constraint satisfaction problems in AI

This module is presented from the official syllabus scope for Introduction to Artificial. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize with constraint satisfaction problems in AI
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition

3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 & conquer approaches, beam stack search applying describe the problem decomposition should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing

techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, techniques - goal trees, ao*, rule based 4 systems applying compare and summarize game playing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

4 algorithms - Minimax, AlphaBeta

4 algorithms - Minimax, AlphaBeta is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 algorithms - Minimax, AlphaBeta using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 algorithms - minimax, alphabeta should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 algorithms - Minimax, AlphaBeta means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 algorithms - Minimax, AlphaBeta □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3 https://onlinecourses.swayam2.ac.in/arp19_ap79/preview

Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3 https://onlinecourses.swayam2.ac.in/arp19_ap79/preview is an official topic in Module 4 of Introduction to Artificial. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, ‘A Guide to Expert Systems’, Pearson Education Online Resources: Sl. No. Website Link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3 https://onlinecourses.swayam2.ac.in/arp19_ap79/preview using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarize constraint satisfaction problem 3 understanding finding optimal paths -branch & bound, divide & conquer approaches problem decomposition: goal trees, ao*, rule based system. game playing: type of games in ai - optimal decisions in games -zero sum game, game tree, minimax algorithm, alphabeta algorithm. constraint satisfaction problems(csp) – definition, example problems, constraint propagation- inference in csps, backtracking search for csps, structure of csp problems. text / reference: t/r book title / author t1 elaine rich and kevin knight. artificial intelligence, tata mcgraw hill stuart russell and peter norvig. artificial intelligence: a modern approach, 3rd t1 edition. prentice hall. r1 deepak khemani. a first course in artificial intelligence, mcgraw hill education (india) r2 stefan edelkamp and stefan schroedl. heuristic search, morgan kaufmann r3 donald a.waterman, ‘a guide to expert systems’, pearson education online resources: sl. no. website link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3 https://onlinecourses.swayam2.ac.in/arp19_ap79/preview should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, ‘A Guide to Expert Systems’, Pearson Education Online Resources: Sl. No. Website Link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3 https://onlinecourses.swayam2.ac.in/arp19_ap79/preview means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, ‘A Guide to Expert Systems’, Pearson Education Online Resources: Sl. No. Website Link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2

https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3
https://onlinecourses.swayam2.ac.in/arp19_ap79/preview □□□□□□□□□□□□ □□□□□
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Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Familiarize with constraint satisfaction problems in AI

            Solve optimal paths - Branch & Bound, Divide                 Applying
  M4.01                                                          3
            & Conquer approaches, Beam Stack Search

Applying
              Describe the Problem Decomposition
  M4.02       techniques - Goal Trees, AO*, Rule Based             4
              Systems

Applying
              Compare and summarize game playing
  M4.03                                                            4
              algorithms - Minimax, AlphaBeta

  M4.04       Familiarize Constraint Satisfaction Problem          3
Understanding

              Series Test – II                                     1

Contents:
Finding optimal paths -Branch & bound, Divide & Conquer approaches
Problem Decomposition: Goal Trees, AO*, Rule Based System.
Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game,
Game
tree, Minimax Algorithm, AlphaBeta Algorithm.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Define AI, foundation and history of AI.. (3 marks)

Answer: Begin with the standard definition or identity of *Define AI, foundation and history of AI.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Discuss applications of AI.. (3 marks)

Answer: Begin with the standard definition or identity of *Discuss applications of AI.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain the Turing test.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the Turing test.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Explain AI problems. (3 marks)

Answer: Begin with the standard definition or identity of *Explain AI problems*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain knowledge representation and reasoning Understanding Describe the representation and reasoning 2. (3 marks)

Answer: Begin with the standard definition or identity of *knowledge representation and reasoning Understanding Describe the representation and reasoning 2*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain about objects, relations, events, actions, time Understanding and space. (3 marks)

Answer: Begin with the standard definition or identity of *about objects, relations, events, actions, time Understanding and space*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Demonstrate Propositional Logic. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate Propositional Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain Demonstrate First Order Logic. (3 marks)

Answer: Begin with the standard definition or identity of *Demonstrate First Order Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Module 3

Define or explain Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth. (3 marks)

Answer: Begin with the standard definition or identity of *Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain 6 Applying First Search Implement Best first search, Hill Climbing, A*. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying First Search Implement Best first search, Hill Climbing, A**. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search. (3 marks)

Answer: Begin with the standard definition or identity of *6 Applying & Beam Search Search Algorithm Terminologies, Types of search algorithms Uninformed Search: Breadth First Search, Depth First Search Informed Search: Heuristic Search - Best first search, Hill Climbing, A* algorithm Beam Search*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition. (3 marks)

Answer: Begin with the standard definition or identity of *3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing. (3 marks)

Answer: Begin with the standard definition or identity of *techniques - Goal Trees, AO*, Rule Based 4 Systems Applying Compare and summarize game playing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 algorithms - Minimax, AlphaBeta. (3 marks)

Answer: Begin with the standard definition or identity of *4 algorithms - Minimax, AlphaBeta*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction Problems(CSP) – Definition, Example Problems, Constraint Propagation-inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3 https://onlinecourses.swayam2.ac.in/arp19_ap79/preview. (3 marks)

Answer: Begin with the standard definition or identity of *Familiarize Constraint Satisfaction Problem 3 Understanding Finding optimal paths -Branch & bound, Divide & Conquer approaches Problem Decomposition: Goal Trees, AO*, Rule Based System. Game Playing: Type of Games in AI - Optimal decisions in games -Zero Sum Game, Game tree, Minimax Algorithm, AlphaBeta Algorithm. Constraint Satisfaction*

Problems(CSP) – Definition, Example Problems, Constraint Propagation- inference in CSPs, Backtracking search for CSPs, Structure of CSP problems. Text / Reference: T/R Book Title / Author T1 Elaine Rich and Kevin Knight. Artificial Intelligence, Tata McGraw Hill Stuart Russell and Peter Norvig. Artificial Intelligence: A Modern Approach, 3rd T1 Edition. Prentice Hall. R1 Deepak Khemani. A First Course in Artificial Intelligence, McGraw Hill Education (India) R2 Stefan Edelkamp and Stefan Schroedl. Heuristic Search, Morgan Kaufmann R3 Donald A.Waterman, 'A Guide to Expert Systems', Pearson Education Online Resources: Sl. No. Website Link 1 https://www.tutorialspoint.com/virtualization2.0/index.htm 2 https://onlinecourses.swayam2.ac.in/aic20_sp06/preview 3 https://onlinecourses.swayam2.ac.in/arp19_ap79/preview. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Define AI, foundation and history of AI.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **knowledge representation and reasoning Understanding Describe the representation and reasoning 2**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Summarise types of search algorithms 2 Understanding Implement Breadth First Search and Depth**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **3 & Conquer approaches, Beam Stack Search Applying Describe the Problem Decomposition**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No. Website Link
- https://www.tutorialspoint.com/virtualization2.0/index.htm
- https://onlinecourses.swayam2.ac.in/aic20_sp06/preview
- https://onlinecourses.swayam2.ac.in/arp19_ap79/preview

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3382. Verify institutional updates against the current official curriculum.